BELIMO

USER MANUAL

Energy Valve

Savings Estimator

How to drive the tool tab by tab: every dropdown, every field you can type in, and what every number it hands back actually means — with particular attention to the financial results, which are the ones most often misread.

Tool version	V.2026.12.0
Workbook	Belimo Energy Valve Estimator Tool.xlsm
Languages	English · Español · Português · Français
Units	US or Metric
Maintained by	Rick Smith — Rick.Smith@us.belimo.com

Belimo Americas · Internal field reference

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1 What the tool does

One sentence: it estimates what a building saves when Energy Valves let the chilled-water plant run at a higher Delta-T.

Every chilled-water plant is sized around a **design Delta-T** — the temperature rise between supply and return. Most plants never achieve it. They run at a lower Delta-T, which means that to deliver the same cooling they must push more water around the building.

That costs money twice over, and the tool separates the two:

SAVINGS STREAM	WHY IT HAPPENS
Pumping savings	Higher Delta-T means less flow for the same load. Pump power falls roughly with the cube of flow, so a modest flow reduction is a large power reduction.
Chiller savings	The same cooling needs less flow, so fewer chillers have to run and their pumps, tower fans and controls stop. The default method credits these auxiliaries; the Chiller tab explains the alternative.

The tool models the plant across six load bins over a full year, prices the energy at your electricity rate, and then puts a financial case around the difference. The tabs run left to right in the order you should fill them in.

The screenshots in this manual show a real worked example - a 1,791 ton, two-chiller plant (the default example, the same one Belimo's original estimator opens with) lifting its Delta-T from 11 °F to 14 °F for a $290,000 incremental investment. Your own numbers will differ; what matters is that every figure shown traces back to inputs you control.

2 Before you start

Language and units

Top-left of the Home tab. Both are dropdowns and both change the entire workbook the moment you pick them — labels, headers, unit suffixes and every converted number, on every tab.

CONTROL	CHOICES	NOTES
Language	English, Español, Português, Français	Translates labels, headings and the written summaries on the Results tab.
Units	US, Metric	Switches every flow, head, temperature, power and volume figure. The calculation is always done in US units and converted for display, so switching back and forth never drifts.
Save Settings	tick box	Ticked, the workbook reopens in the language and units you left it in. Unticked, it resets to English / US on every open.

Which cells can I type in?

The workbook tells you, and the convention is the same on every tab. The Energy, Chiller and Pump tabs print the key on the sheet itself:

SHADING	MEANING
Pale / near-white	Editable field. Type here.
Blue-grey	Label or calculated result. Protected — the tool writes it.
Orange band	Section heading.

Sheets are protected so a stray keystroke cannot break a formula. If a cell will not accept typing, that is the tool telling you it is a result, not an input.

About screen and updates

Click the round blue (i) button at the top right of the Home tab, beside the red PDF Download button, to open the About screen. It shows the tool's name, its version and last-updated date, the internal-use notice, and who maintains the tool. It also has three buttons and a Close button.

The About screen, opened from the blue (i) button on the Home tab.

BUTTON	WHAT IT DOES
Check for Updates	Asks the update source whether a newer release exists. If one does, you are told what is new and may choose to download it.
Download Fresh Copy	Downloads the current published release straight away, even if you are already up to date – useful if you think your copy is damaged.
Download User Manual	Saves the current published copy of this manual (a PDF) to a folder you choose. If none has been published yet, the tool tells you so.
Close	Closes the About screen.

Each time the workbook opens it quietly checks Belimo's update source for a newer release. When one is available you are offered a summary of what changed. If you choose to install it, you pick where the new copy is saved through a normal Windows Save As dialog, just like any other file – your current copy is never changed or deleted. If the update source cannot be reached (for example you are offline), the check does nothing and the tool works as normal.

3 Home tab

The only tab most jobs need. Plant inputs at the top, the Delta-T comparison table in the middle, headline financials and greenhouse-gas equivalents below.

BELIMO

Home | Energy | Chiller | Pump | F

Belimo Energy Valve™ Savings Estimator

Language	English	Units	US	☐ Save Settings	
Project Information	Enter Project Name				
Climate region	North East	Chiller cooling	Water Cooled	Plant design	Primary/Secondary
Utility Company	None	Program			

INPUT DATA: CHILLED WATER PLANT

	Value	Unit
Electricity cost (average)	0.1701	$/kWh
Design cooling load	1791	tons
Number of chillers	2	pcs.
Operating hours per day	12	hours
Operating days per week	5	days
Chiller design efficiency	0.6	kW/ton
Actual Secondary Pump Head (Design)	135	ft
Minimum distribution pump flow percentage	30	%
Pump, VFD and motor efficiency	65	%
Design Delta-T	14	°F
Actual Delta-T before Energy Valve	11	°F
Future Energy Valve Delta-T setting	14	°F

ANNUAL ENERGY COSTS

Delta-T [°F]	Avg [GPM]	Avg [tons]	Pumping Cost	Chiller Cost	Total Cost	Energy [kWh]
10	2505	1461	$ 37,110	$ 96,052	$ 133,163	782,850
11	2412	1407	$ 34,259	$ 94,006	$ 128,266	754,060
12	2252	1313	$ 28,098	$ 87,247	$ 115,345	678,102
13	2097	1223	$ 22,898	$ 83,881	$ 106,779	627,745
14	1965	1146	$ 19,071	$ 80,996	$ 100,067	588,283
15	1840	1073	$ 15,666	$ 78,272	$ 93,937	552,248
Annual Energy Savings for			44%	14%	22%	%
changing Delta T from 11 to 14			$15,188	$13,011	$28,199	(165,777)

OTHER SAVINGS TO CONSIDER

	Value	Unit
Cooling tower water treatment chemicals savings (reduced runtime)	600	$/year
Reduced runtime of lag chillers, extending the life of chillers and pumps	0	$/year
Additional cooling loads (GPM) added without upsizing pumps	0	$/year

INPUT FINANCIALS

	Value	Unit
Incremental Energy Valve investment	290,000	$
Discount rate	10	%
Annual energy-cost inflation	3	%

FINANCIAL ANALYSIS RESULTS

Investment Horizon	5 years	10 years	15 years	20 years
Savings-Investment Ratio (SIR)	0.40	0.68	0.89	1.04
Net Present Value (NPV)	($174,730)	($91,757)	($32,031)	$10,961

GREENHOUSE GAS EQUIVALENTS

	Value	Unit
Annual carbon dioxide avoided	245,483	pounds
Tree seedlings grown for 10 years	1,856	tree seedlings
Equivalent annual home electricity use	14	homes

DISCLAIMER

The Home tab end to end. Everything else in the workbook is downstream of this page.

Project identification

FIELD	CELL	WHAT TO ENTER
Project	F7	Free text. Job or building name — appears on printouts.
Information	F8	Free text. A one-line description, e.g. “4 chillers; installed 1600 tons”.
Climate region	F9	Dropdown. Picks the cooling-degree-day profile from the Energy tab.
Chiller cooling	J9	Dropdown — Water Cooled or Air Cooled.
Plant design	N9	Dropdown — Primary/Secondary or Variable Primary.
Utility Company	F10	Dropdown. Choosing a utility with a supported programme reveals its rebate tab.
Program	J10	Dropdown. Populated from the utility above. ConEd’s C&I programme adds the two ConEd tabs.

Input data: chilled water plant

FIELD	CELL	UNIT	WHAT IT IS
Electricity cost (average)	L12	$/kWh	Blended all-in rate. Everything priced in dollars runs through this one number.
Design cooling load	L13	tons	Plant capacity at design conditions.
Number of chillers	L14	pcs.	How many machines share the load.
Operating hours per day	L15	hours	With days per week and the weeks per year on the Energy tab, sets annual run hours.
Operating days per week	L16	days	
Chiller design efficiency	L17	kW/ton	Full-load kW/ton from the chiller submittal — not IPLV. Becomes KW1 on the Chiller tab.
Actual Secondary Pump Head	L18	ft	Head the distribution pumps work against at design flow.
Minimum distribution pump flow	L19	%	The flow the pumps still deliver at very low load, as a percentage of design flow — set by any bypass loop, 3-way valves or the minimum pump speed. Default 30%. No load bin runs below it.
Pump, VFD and motor efficiency	L20	%	Combined wire-to-water efficiency.
Design Delta-T	L21	°F	What the plant was designed to achieve.
Actual Delta-T before Energy Valve	L22	°F	What it actually runs at today — the “before” case.
Future Energy Valve Delta-T setting	L23	°F	What you expect to hold with Energy Valves — the “after” case.

The three Delta-T fields are the heart of the estimate. Actual and Future are the two cases the whole comparison is built from, and the tool enforces a gap between them so the pair always makes physical sense.

Annual energy costs

Calculated — nothing to type. One row per Delta-T from your Actual up through your Future setting, colour-graded red (worst) to green (best): average flow, average tonnage, pumping cost, chiller cost, total cost and annual energy. The summary line beneath is the number most customers care about — the percentage and dollar reduction for moving from the Actual Delta-T to the Future one.

Other savings to consider

Three optional yearly dollar values that are added to the energy savings on the Financials tab (row Maintenance/Other Savings) and grow with the energy-cost inflation rate. Enter 0 for anything that does not apply.

FIELD	CELL	UNIT	WHAT IT IS
Cooling tower water treatment chemicals savings (reduced runtime)	L36	$/year	Water treatment saved when the tower runs fewer hours; water-cooled plants only. A rough estimate is $300 per cooling tower per year. Default $600; enter 0 for air-cooled plants.
Reduced runtime of lag chillers, extending the life of chillers and pumps	L37	$/year	The yearly value of postponing a chiller or pump replacement; it depends on replacement cost, age and remaining life. Default 0.
Additional cooling loads (GPM) added without upsizing pumps	L38	$/year	The yearly value of being able to add loads to the secondary loop without replacing distribution pumps. Default 0.

Input financials

FIELD	CELL	UNIT	WHAT IT IS
Incremental Energy Valve investment	L41	$	Incremental is the operative word — the extra cost of Energy Valves over the valves that would have been fitted anyway, not the whole valve package.
Discount rate	L42	%	Feeds every present-value calculation. See section 7.3.
Annual energy-cost inflation	L43	%	Escalates the savings year on year, because energy prices rise.

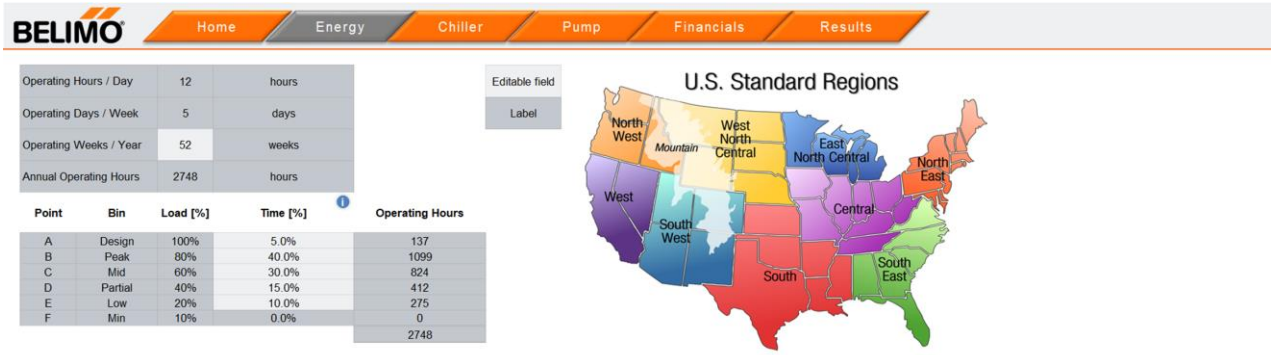
Financial analysis results and greenhouse gas equivalents

The **Savings-Investment Ratio (SIR)** and **Net Present Value (NPV)** at 5, 10, 15 and 20 years, colour-graded, with the full workings on the Financials tab. Both are explained properly in section 7.3 - for now, treat SIR as how many dollars back per dollar in, and NPV as what the project is worth in today's money.

Below them the emissions avoided, expressed as pounds of CO_2 and as the two equivalents most people can picture - tree seedlings grown for ten years, and homes' annual electricity use. They are calculated from the kilowatt-hours saved, using U.S. EPA factors (see section 11).

4 Energy tab

Where the year gets divided into load bins, and where the climate profile lives.



Operating hours and weeks, the six load bins, and the regions map. The two grey boxes are the on-sheet key for editable versus label cells.

Load bins

Rather than simulate 8,760 hours, the tool uses six representative operating points — the same six as Belimo's original estimator — and assigns each a share of the year (default 5 / 40 / 30 / 15 / 10 / 0%). Only the Time [%] column is editable; the last point takes whatever is left, so the six shares always total 100%. Operating Weeks / Year (default 52) is the one schedule input that lives on this tab. Annual Operating Hours is your schedule (hours per day × days per week × weeks per year) multiplied by the hours factor of your region, which is never more than 1.00 — so a plant scheduled 24 hours a day, 7 days a week for 52 weeks can never show more than 8,736 hours. The Results tab reports the hours used, the average load and the equivalent full-load hours in a Basis line under the headline savings.

POINT	BIN	LOAD	CELL	MEANING
A	Design	100%	E10	Full-load hours — the design day.
B	Peak	80%	E11	Hot-weather part load.
C	Mid	60%	E12	Typical warm-weather operation.
D	Partial	40%	E13	Mild-weather part load.
E	Low	20%	E14	Cool-weather and light-load operation.
F	Min	10%	—	Remainder — calculated so the six shares total 100%.

Change these only with a reason. The defaults are Belimo's standard six-point profile, a weighted mix of common building types. If you have real trend data or a load profile from the BAS, use it — otherwise leave them alone, because the bin split drives every hour, kWh and dollar downstream.

Climate regions

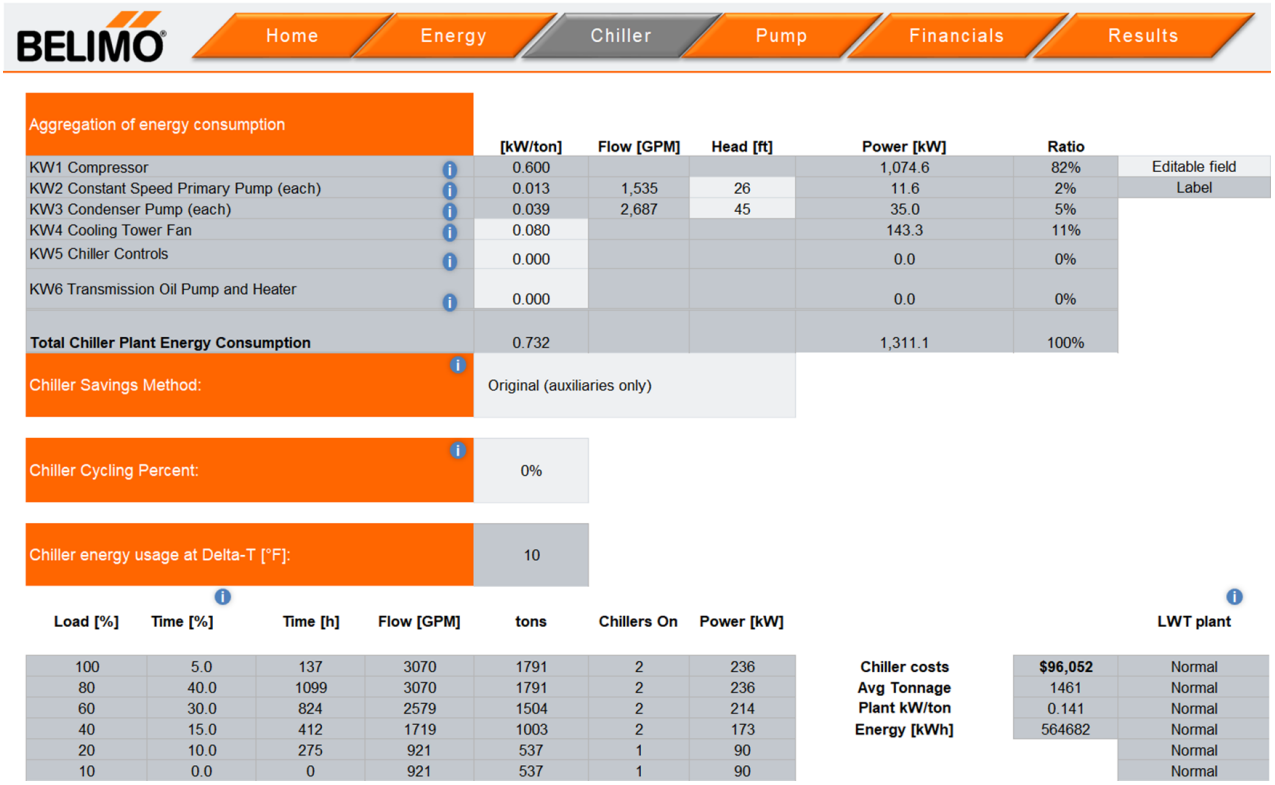
Climate region	Cooling Degree Days [°F·day]	Humidity Adjustment	Hours factor
North East	1034	30%	0.88
Central	1090	30%	0.93
East North Central	975	30%	0.83
West North Central	1123	15%	0.85
South East	2264	30%	1.00
South	2004	30%	1.00
South West	2750	5%	1.00
Mountain	1450	5%	1.00
North West	720	5%	0.50
West	1850	5%	1.00
Custom Region	1526	0%	1.00
Average US	1526	19%	1.00
		Selected Value	0.88

Each region carries cooling degree days, a humidity adjustment and the resulting hours factor: the region's degree days compared with the US average, allowing for humidity, and never more than 1.00 — climate can reduce the operating hours but never adds hours beyond your schedule. The Selected Value at the bottom is the one your Home-tab region choice pulled through.

ROW	CELLS	USE
Custom Region	E33, F33	Editable. Enter your own cooling degree days and humidity adjustment for a site that does not fit a standard region — then select "Custom Region" on the Home tab. It starts neutral (the US-average degree days, factor 1.00); degree days may be 0–6,000 and humidity 0.01–30%.
Average US	—	Reference only — the national average, for sanity-checking.

5 Chiller tab

Breaks total plant efficiency into its six components, then works out chiller energy at each Delta-T.



The kW/ton build-up, the savings method, the cycling allowance, and the first of several per-Delta-T calculation blocks.

Aggregation of energy consumption

Plant efficiency is not just the compressor. This table adds the auxiliaries to get a true plant kW/ton, and shows each as a share of the total. The two pumps are worked out from a flow and a head, so the table also shows their flow, head and power.

COMPONENT	CELL	SOURCE
KW1 Compressor	—	Calculated — your Chiller design efficiency (Home L17). Change it there.
KW2 Constant Speed Primary Pump	H7	Flow calculated, head editable (default 26 ft). Primary/secondary plants only.
KW3 Condenser Pump	H8	Flow calculated (3 GPM per ton), head editable (default 45 ft). Water-cooled plants only.
KW4 Cooling Tower Fan	F9	Editable. Default 0.080 kW/ton. Water-cooled plants only.
KW5 Chiller Controls	F10	Editable. Default 0 kW/ton.
KW6 Transmission Oil Pump and Heater	F11	Editable. Default 0 kW/ton.
Total Chiller Plant Energy Consumption	—	The sum — the plant kW/ton actually used in the costing.

What KW1–KW6 mean and where to get them

The total is multiplied by the tons of chiller capacity running in each load bin, so every figure has to be on the same basis: kilowatts drawn per ton of chiller capacity while that chiller runs. Divide equipment dedicated to one chiller by that chiller's rated tons, and plant-wide equipment by plant capacity. The kilowatts come from the motor nameplate (HP × 0.746 × load factor ÷ motor efficiency, with a real load factor of 0.75–0.85), the pump schedule (GPM × head in ft ÷ (5,308 × overall efficiency)), or a measurement (volts × amps × 1.732 × power factor ÷ 1,000, or a power meter or VFD display).

KW1 Compressor. Electricity the chiller draws per ton at full load — the largest term by far. Take it from the chiller submittal or AHRI certificate (full-load kW/ton), or input kW ÷ rated tons from the nameplate; on site, a chiller kW trend ÷ tons, where tons = GPM × ΔT ÷ 24. Never IPLV: it is a part-load average and understates energy.

KW2 Constant-speed primary pump. The pumps pushing water through each chiller on a primary/secondary plant. The flow of one pump is calculated — design tons × 12,000 ÷ (chillers × 500 × design Delta-T) — and you enter the head (H7, default 26 ft, typical for a primary loop). One pump runs for every chiller that is on: kW = 0.746 × GPM × ft ÷ (3,960 × the pump efficiency on Home). On the default plant 1,535 GPM at 26 ft and 65% gives 11.6 kW per pump, shown as 0.013 kW/ton across the plant. To sanity-check, divide a primary pump's motor kW by the rated tons of the chiller it serves.

KW3 Condenser pump. Moves condenser water between the chiller and the cooling tower; water-cooled plants only. The flow of one pump is calculated at 3 GPM per ton of chiller and you enter the head (H8, default 45 ft). One pump runs for every chiller that is on: on the default plant 2,687 GPM at 45 ft and 65% gives 35.0 kW per pump, shown as 0.039 kW/ton across the plant. Use the condenser pump schedule (GPM and head) or the motor nameplate where you have them.

KW4 Cooling tower fan. The fan motors that reject the heat; water-cooled plants only. From the tower submittal or nameplates — total every fan, convert to kW, ÷ the tons the tower serves. Typically 0.02–0.08, with fixed- or two-speed fans at the high end; the default is 0.08. The tool scales this with the tons the chillers deliver, not with fan speed, so it cannot credit a tower VFD.

KW5 Chiller controls. The control panel, sensors, actuators and starter electronics. From the control-panel electrical data on the submittal — usually 0.5–2 kW per chiller — ÷ rated tons. Typically 0.002–0.005; it rarely matters. The default is 0, because a manufacturer's full-load rating usually already includes it.

KW6 Transmission oil pump and heater. The oil pump that lubricates the compressor bearings and, on gear-driven centrifugals, the gearbox — plus the oil sump heater. From the auxiliary electrical data on the submittal: oil pump motor (often 1–3 HP) plus heater (often 1–3 kW), ÷ rated tons. Oil-free chillers with magnetic bearings have neither: enter 0. The default is 0 (a full-load rating usually includes it).

Double counting. A manufacturer's full-load rating usually covers the whole chiller package, often including its controls and oil pump. Check the submittal footnotes — if they are included, KW5 and KW6 overlap KW1 slightly, about 2%.

Easy to inflate. With Original, the default, KW2–KW6 are the whole chiller-side saving. On the shipped example every +0.01 kW/ton on KW4 adds about 9% to it and on KW5 or KW6 about 5% (under Full plant, about 1.4%); every +10 ft on the condenser pump head adds about 4% and on the primary pump head about 2%. Leave the defaults unless you have submittal data or measurements.

Chiller Savings Method

Editable dropdown (F13, default Original). Decides how much of the chiller plant's energy the better Delta-T is credited with. The pump saving is the same either way.

OPTION	WHAT IS CREDITED	USE IT WHEN
Original (auxiliaries only)	The method of the original 2021 estimator. The building still needs the same tons, so compressor energy is not counted — only the primary and condenser pumps, tower fans, controls and oil pumps that stop with fewer chillers running.	The default. The figure to put in front of a customer or a utility — the next section explains why.
Full plant (incl. compressor)	Every chiller the plant no longer needs to run is credited its full energy — compressor included, at the full-load kW/ton from Home — plus its pumps, tower fans, controls and oil pump.	Only as an upper bound, and say so. Most of its extra saving is compressor energy that is not really saved.

It moves the answer a lot. On the shipped example the total saving is $28,199 a year with Original and $104,339 with Full plant — the chiller side is $13,011 against $89,151, while the pump side stays at $15,188. Say which method a figure uses whenever you quote it.

Original follows the 2021 formula: on the same inputs the tool gives the same figure to the dollar. Two details are handled slightly differently where they make no difference at the defaults: tower-fan savings count only for water-cooled plants, and KW5 and KW6 are multiplied by tons rather than by the number of chillers (both default to 0). Like every field, the dropdown has a blue i button that explains both options in the selected language.

Compressor versus auxiliaries — why Original is the default

Two kinds of equipment use electricity in a chiller plant, and a better Delta-T affects them very differently.

The compressor is the refrigeration machine inside each chiller — where the cooling is actually made. Its power follows the tons of cooling the building needs at that moment. It is KW1: 0.600 kW/ton on the example, 82% of plant power.

The auxiliaries are the supporting equipment that runs because a chiller is switched on, however hard that chiller is working — KW2–KW6.

TERM	EQUIPMENT	KW/TON	SHARE
KW2	Constant-speed primary pump (one per chiller)	0.013	2%
KW3	Condenser pump	0.039	5%
KW4	Cooling tower fans	0.080	11%
KW5	Chiller controls	0.000	0%
KW6	Oil pump and oil heater	0.000	0%
Total	Auxiliaries	0.132	18%

What happens when Delta-T improves. Take an hour when the building needs 895 tons (half of the example plant's 1,791 tons), and each chiller is rated 895.5 tons, designed for 1,535 GPM. At a Delta-T of 11 °F the plant has to push 1,954 GPM, more than one chiller can supply, so both chillers run. At the design Delta-T of 14 °F the same 895 tons needs 1,535 GPM, so one chiller runs.

POWER DRAWN	2 CHILLERS ON	1 CHILLER ON	SAVING
Auxiliaries (0.132 kW/ton × 895.5 t per chiller)	236 kW	118 kW	118 kW — real. The pumps, fan and heater of the idle chiller stop.
Compressors, as Full plant counts them	1,075 kW	537 kW	537 kW
Compressors, physically	about 537 kW	about 537 kW	About zero — the same 895 tons is made either way.

Full plant charges every running chiller its full-load compressor power (2 × 895.5 × 0.60 = 1,075 kW). But the building still needs only 895 tons, so the compressors make 895 tons whichever machines run: about 895.5 × 0.60 = 537 kW. Switching one chiller off moves that load onto the other; it does not make it go away.

Why Original is the default. Most of the Full plant saving is compressor energy that is not really saved. Original gives the figure an engineer can defend, and it is the one to put in front of a customer or a utility. Some compressor effect is real — a chiller idling at very low load often runs at a worse kW/ton — but it is a few percent of compressor energy, it can go either way on variable-speed chillers, and it is nowhere near half.

The tables above illustrate the principle rather than reproduce the tool's exact arithmetic — the tool, for instance, scales cooling-tower fans with water flow rather than with chillers running.

Chiller Cycling Percent

Editable (F15, default 0%). An allowance for the efficiency penalty of chillers short-cycling at low load. Raise it if the plant is known to cycle badly.

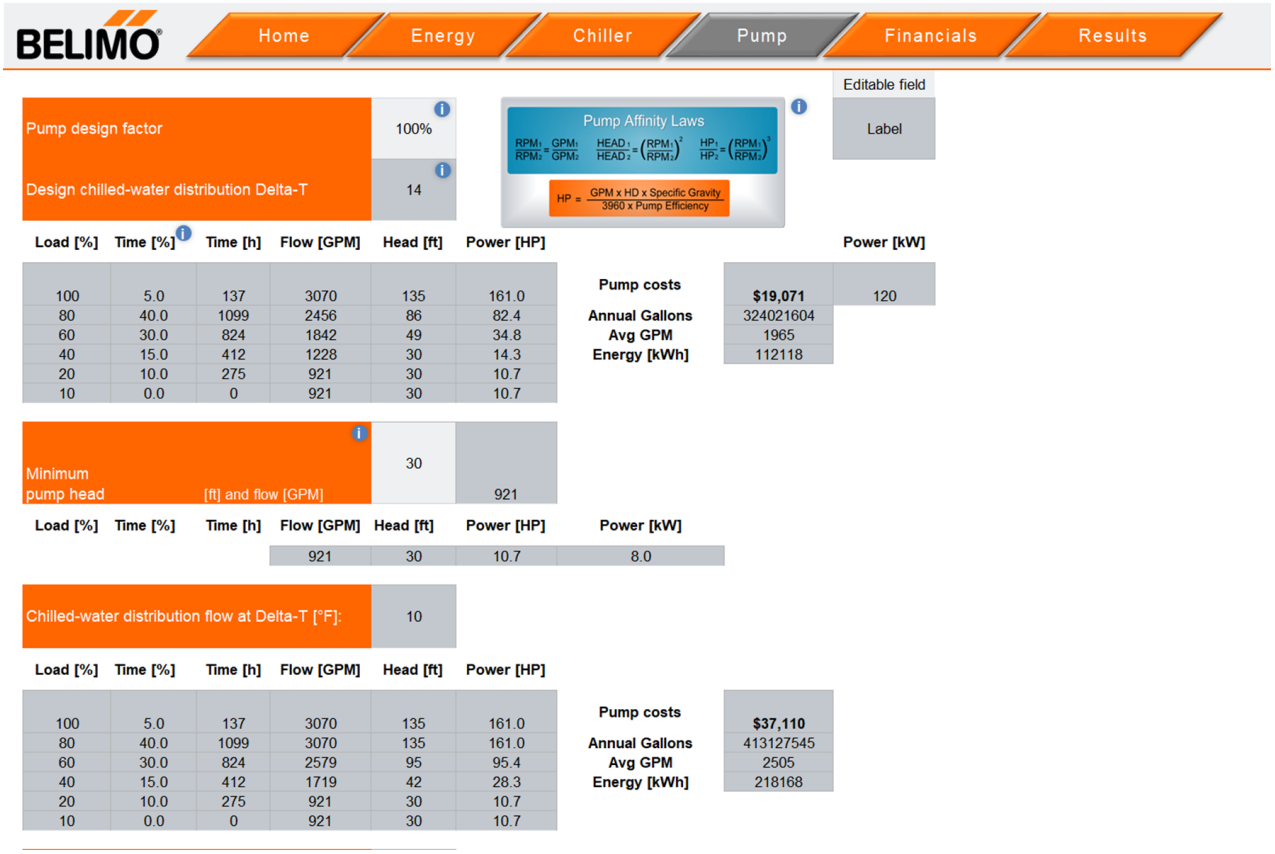
Per-Delta-T blocks

Below that, one block per Delta-T in the comparison range. Each is headed by its Delta-T and shows the six bins with load, time, flow, tons, how many chillers are running and the resulting power — then chiller cost, average tonnage, plant kW/ton and annual kWh. All calculated. On primary/secondary plants an LWT plant column flags any load where the secondary-loop flow is larger than the primary pumps can supply.

About the tons figures. The tons shown here — and the average tonnage on the Home tab and the average cooling load on the Results tab — are the flow multiplied by the design Delta-T and divided by 24, never more than the design load. That is the convention of Belimo’s original estimator: below the design Delta-T the extra flow counts as extra tonnage, and the cooling-tower fan and the plant kW/ton follow it.

6 Pump tab

The same idea for the distribution pumps — flow, head and power in each bin, at each Delta-T.



Design case at the top, then a block per Delta-T. Pump costs, annual gallons, average GPM and energy sit to the right of each block.

FIELD	CELL	WHAT IT IS
Pump design factor	F5	Editable, default 100%. A safety/oversizing multiplier on design flow. Leave at 100% unless you know the pumps were deliberately oversized.
Minimum pump head	F15	Editable, default 30 ft. A floor on head as flow falls. Real systems never reach zero head — static lift, control valves and minimum-speed limits keep it above zero, and without a floor the affinity laws would overstate savings at low load.
Minimum flow	G15	Calculated. The minimum distribution pump flow percentage on the Home tab (default 30%) times the design flow. No load bin runs below it.
Bin tables	—	Calculated. Load, time, flow, head and horsepower per bin, plus cost, annual gallons, average GPM and kWh.

Why pumping savings look large. Pump power varies with roughly the cube of flow. Cutting flow by a third cuts pump power by about 70%. That is why the pumping percentage saving on the Home tab is usually much bigger than the chiller percentage — though in absolute dollars the pumping and chiller savings are usually of a similar size — which is larger depends on the plant.

7 Financials tab

The full ten-year cash-flow model behind the headline numbers on Home. This is the tab that gets misread most often, so it is worth understanding properly.

FINANCIAL ANALYSIS OF EFFICIENCY IMPROVEMENTS
Simulating Streams of Cash Inflows and Outflows

Discount rate:	10%
Finance rate	10%
Reinvestment rate	10%
Inflation Rate	3%

	Today	End of Year 1	End of Year 2	End of Year 3	End of Year 4	End of Year 5	End of Year 6	End of Year 7	End of Year 8	End of Year 9	End of Year 10
Date:	0	1	2	3	4	5	6	7	8	9	10
CASH OUTFLOWS											
Single Investment	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Phased Investment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Financed Investment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Subtotal	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Rebate Rec'd at Project Start	$0										
Subtotal Outflows	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
CASH INFLOWS											
Energy savings		$28,199	$29,045	$29,916	$30,814	$31,738	$32,690	$33,671	$34,681	$35,721	$36,793
Maintenance/Other Savings		$600	$618	$637	$656	$675	$696	$716	$738	$760	$783
Rebates Rec'd after Project Start		$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Subtotal Inflows		$28,799	$29,663	$30,553	$31,469	$32,413	$33,386	$34,387	$35,419	$36,481	$37,576
Annual cash flow	($290,000)	$28,799	$29,663	$30,553	$31,469	$32,413	$33,386	$34,387	$35,419	$36,481	$37,576
PV of Outflows (for SIR)	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
PV of Inflows (for SIR)	$0	$26,181	$24,515	$22,955	$21,494	$20,126	$18,845	$17,646	$16,523	$15,472	$14,487
Annual Present Value	($290,000)	$26,181	$24,515	$22,955	$21,494	$20,126	$18,845	$17,646	$16,523	$15,472	$14,487

Note: Certain Returns Vary Depending on the Length of the Analysis Term

	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV)	($91,757)	($263,819)	($239,305)	($216,350)	($194,856)	($174,730)	($155,885)	($138,239)	($121,716)	($106,244)	($91,757)
Simple Payback Period (SPP)	10.1										
Return on Investment (ROI)	9.9%										
Internal Rate of Return (IRR)	2.3%	-90.1%	-62.7%	-41.6%	-27.5%	-17.9%	-11.1%	-6.2%	-2.6%	0.2%	2.3%
Mod. Internal Rate of Return (MIRR)	5.9%	-90.1%	-54.0%	-30.3%	-16.7%	-8.5%	-3.3%	0.3%	2.8%	4.6%	5.9%
Savings-Investment Ratio (SIR)	0.7	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.6	0.6	0.7
	V.2026.12.0										

The whole tab: rates top-left, cash flows across ten years, then the metric block at the bottom.

7.1 The four rates

RATE	CELL	SET WHERE	WHAT IT DOES
Discount rate	B9	Home L42	Converts future dollars into today's dollars. Usually the organisation's cost of capital or required return. Raising it lowers both headline results, because distant savings count for less.
Finance rate	B10	Editable here	What it costs to borrow the money. Used only by the Modified Internal Rate of Return (section 7.3).
Reinvestment rate	B11	Editable here	What the savings earn once received. Used only by the Modified Internal Rate of Return.
Inflation rate	B12	Home L43	Escalates the annual energy saving. This is why the year-10 saving is larger than the year-1 saving.

7.2 The cash-flow grid

	Today	End of Year 1	End of Year 2	End of Year 3	End of Year 4	End of Year 5	End of Year 6	End of Year 7	End of Year 8	End of Year 9	End of Year 10
Date:	0	1	2	3	4	5	6	7	8	9	10
	CASH OUTFLOWS										
Single Investment	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Phased Investment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Financed Investment	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Subtotal	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Rebate Rec'd at Project Start	$0										
Subtotal Outflows	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
	CASH INFLOWS										
Energy savings		$28,199	$29,045	$29,916	$30,814	$31,738	$32,690	$33,671	$34,681	$35,721	$36,793
Maintenance/Other Savings		$600	$618	$637	$656	$675	$696	$716	$738	$760	$783
Rebates Rec'd after Project Start		$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Subtotal Inflows		$28,799	$29,663	$30,553	$31,469	$32,413	$33,386	$34,387	$35,419	$36,481	$37,576
Annual cash flow	($290,000)	$28,799	$29,663	$30,553	$31,469	$32,413	$33,386	$34,387	$35,419	$36,481	$37,576
PV of Outflows (for SIR)	($290,000)	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
PV of Inflows (for SIR)	$0	$26,181	$24,515	$22,955	$21,494	$20,126	$18,845	$17,646	$16,523	$15,472	$14,487
Annual Present Value	($290,000)	$26,181	$24,515	$22,955	$21,494	$20,126	$18,845	$17,646	$16,523	$15,472	$14,487

Columns run Today, then End of Year 1 through End of Year 10. Money out is negative; money in is positive.

ROW	EDITABLE?	MEANING
Single Investment	Years 1–10	The up-front cost. Today's column comes straight from Home L41; later columns let you add a further lump.
Phased Investment	All columns	For work spread over several years — a floor a year, say.
Financed Investment	All columns	For a financed package, entered as the payments fall.
Rebate Rec'd at Project Start	Today only	Incentive money in hand at day zero. Reduces the net outflow.
Subtotal Outflows	Calculated	All investment, less any start rebate.
Energy savings	Calculated	From the plant model, escalated each year by the inflation rate.
Maintenance/Other Savings	Calculated	The three Other savings inputs on the Home tab added together, escalated each year by the inflation rate.
Rebates Rec'd after Project Start	Years 1–10	Incentives paid later, e.g. on measured performance.
Subtotal Inflows	Calculated	Everything coming back.
Annual cash flow	Calculated	Inflows plus outflows for that year. The series every metric is built from.
PV of Outflows (for SIR) / PV of Inflows (for SIR)	Calculated	Each side discounted to today. Used for the Savings-Investment Ratio. The Outflows row carries an i button explaining present value.
Cumulative Cash Flows	Calculated	Running total, undiscounted. The year it turns positive is the true payback year.
Annual Present Value	Calculated	That year's cash flow in today's money.
Cumulative Present Value	Calculated	Running total of the above - the last figure is the Net Present Value.

7.3 What the terms mean

	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV)	($91,757)	($263,819)	($239,305)	($216,350)	($194,856)	($174,730)	($155,885)	($138,239)	($121,716)	($106,244)	($91,757)
Simple Payback Period (SPP)	10.1										
Return on Investment (ROI)	9.9%										
Internal Rate of Return (IRR)	2.3%	-90.1%	-62.7%	-41.6%	-27.5%	-17.9%	-11.1%	-6.2%	-2.6%	0.2%	2.3%
Mod. Internal Rate of Return (MIRR)	5.9%	-90.1%	-54.0%	-30.3%	-16.7%	-8.5%	-3.3%	0.3%	2.8%	4.6%	5.9%
Savings-Investment Ratio (SIR)	0.7	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.6	0.6	0.7
	V.2026.12.0										

The six metrics, each shown at every horizon from one to ten years.

PV

Present Value

What a future amount of money is worth **today**. A dollar saved in year eight is not worth a dollar now — you could have invested that dollar in the meantime. Present value shrinks each future amount back accordingly.

How the tool gets it: cash flow ÷ (1 + discount rate)^years. Everything else on this tab is built from it.

NPV

Net Present Value

Add up the present value of every year — including the money spent on day one, which is negative — and you have the NPV. It is the single best answer to "is this project worth doing?"

NPV > 0 → worth doing. NPV < 0 → not at this discount rate.

Positive means the project returns more than the discount rate you demanded, and the NPV is the surplus, in today's dollars. Negative does not mean it loses money — it means it does not clear the bar you set.

How the tool gets it: the sum of the Annual Present Value row across Today plus all ten years.

SIR

Savings-Investment Ratio

The present value of everything coming in, divided by the present value of everything going out. A ratio rather than an amount, so it compares projects of different sizes on equal terms.

SIR > 1.0 → returns more than it costs. SIR < 1.0 → does not pay back in present-value terms.

An SIR of 1.8 means every dollar invested returns $1.80 in today's money. Many public-sector and utility incentive programmes require a minimum SIR, which is why it appears on the Home tab alongside NPV.

How the tool gets it: PV of Inflows ÷ |PV of Outflows|, both summed across the horizon.

SPP

Simple Payback Period

How many years of savings it takes to repay the investment, ignoring the time value of money entirely. The number everyone asks for first.

How the tool gets it: investment ÷ first-year cash flow. Note it uses the first year only — because savings escalate with inflation, the real payback usually arrives slightly sooner than SPP suggests. Use the Cumulative Cash Flows row to see the actual crossover year.

ROI

Return on Investment

Here, the **first-year** return: what the project returns in year one as a percentage of what it cost.

How the tool gets it: year-1 cash flow ÷ investment. This is NOT a lifetime return, which is what people usually assume the term means. In the worked example it reads 9.9% - meaning year one alone returns about 10 cents on every dollar invested, not that the project earns 9.9% a year for its whole life. Quote it carefully, or quote the Internal Rate of Return instead.

IRR

Internal Rate of Return

The discount rate at which the NPV would come out exactly zero — in effect the interest rate the project itself earns. Compare it against your hurdle rate: if the IRR beats the rate you could get elsewhere, the project is attractive.

How the tool gets it: Excel's IRR over the annual cash flow row. Its weakness: IRR quietly assumes every dollar of savings is reinvested at the IRR itself, which is rarely true — which is exactly what MIRR fixes.

MIRR

Modified Internal Rate of Return

On the sheet this row reads "Mod. Internal Rate of Return (MIRR)" — abbreviated only so the label and its i button both fit the column.

IRR's more honest sibling. It separates the two rates IRR conflates: what borrowing costs you, and what the returning savings actually earn. Usually the more defensible figure to put in front of a finance team.

How the tool gets it: Excel's MIRR over the annual cash flow row, using the Finance rate (B10) and Reinvestment rate (B11) you set at the top of this tab. Set those two honestly and MIRR becomes the number worth arguing from.

Which one should you lead with? For a building owner, simple payback and NPV. For a finance department, NPV and MIRR. For a utility or public-sector programme, SIR — it is often written into the eligibility rules.

7.4 Reading the horizon columns

Every metric is repeated across columns headed 1 Year through 10 Years, with a summary figure at the left. Each column answers the same question over a different holding period: if we judged this project over N years, how would it look?

Short horizons look poor and long horizons look better, because the investment lands entirely in year zero while the savings accumulate. That is expected. Quote the horizon that matches how the customer actually evaluates capital — and say which horizon you are quoting.

The worked example, read across. A $290,000 investment against $28,799 of yearly savings (the $28,199 energy savings plus $600 of other savings) takes 10.1 years to pay back. The one-year column shows a Net Present Value of -$263,819, five years -$174,730 and ten years -$91,757; the Home tab shows it reaching -$32,031 at 15 years and turning positive at 20 years (+$10,961). That crossover from negative to positive is exactly what the columns are there to reveal — so quote a horizon the customer actually uses.

7.5 Built-in help - the blue i buttons

Every term on this tab now explains itself. You do not need this manual open to answer "what does that number mean?" in front of a customer.

Each metric row carries a small blue i button just after its label. Click it and a plain-language explanation opens in whichever language the tool is set to. The explanations are written for someone who does not work in finance, and each one closes with a worked example built from your figures rather than a generic illustration - so what the customer reads always matches what is on screen.

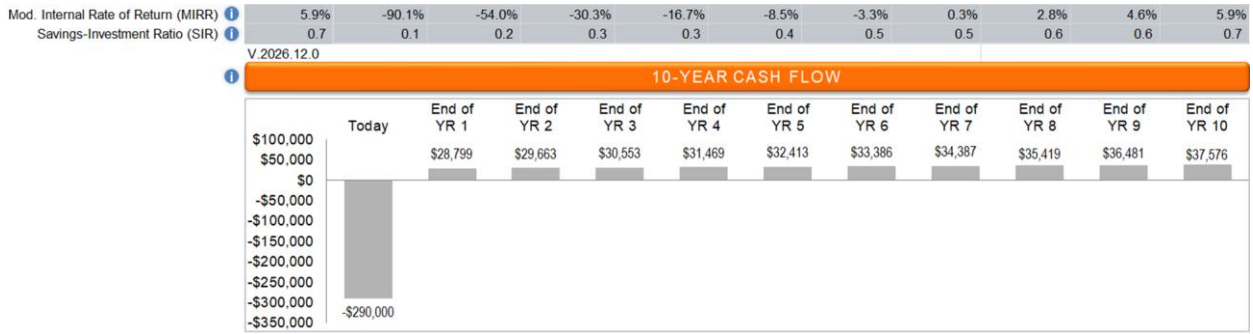
	10 Year	1 Year	2 Years	3 Years	4 Years	5 Years	6 Years	7 Years	8 Years	9 Years	10 Years
Net Present Value (NPV)	($91,757)	($263,819)	($239,305)	($216,350)	($194,856)	($174,730)	($155,885)	($138,239)	($121,716)	($106,244)	($91,757)
Simple Payback Period (SPP)	10.1										
Return on Investment (ROI)	9.9%										
Internal Rate of Return (IRR)	2.3%	-90.1%	-62.7%	-41.6%	-27.5%	-17.9%	-11.1%	-6.2%	-2.6%	0.2%	2.3%
Mod. Internal Rate of Return (MIRR)	5.9%	-90.1%	-54.0%	-30.3%	-16.7%	-8.5%	-3.3%	0.3%	2.8%	4.6%	5.9%
Savings-Investment Ratio (SIR)	0.7	0.1	0.2	0.3	0.3	0.4	0.5	0.5	0.6	0.6	0.7

V.2026.12.0

One button per metric, lined up down the label column.

BUTTON	WHAT IT EXPLAINS
PV of Outflows (for SIR)	Present value - why a dollar received in five years is worth less than a dollar today, and how the discount rate converts one into the other.
Net Present Value (NPV)	What the project is worth today after subtracting the investment, and what the horizon columns are doing.
Simple Payback Period (SPP)	How long the savings take to repay the spend - and, importantly, what SPP ignores.
Return on Investment (ROI)	Total gain as a percentage of the spend, and why timing is invisible to it.
Internal Rate of Return (IRR)	The effective annual return, how to compare it with your cost of money, and why it flatters a project.
Mod. Internal Rate of Return (MIRR)	Why MIRR is the more defensible figure, and which two rates it uses.
Savings-Investment Ratio (SIR)	Dollars saved per dollar invested, with the 1.0 break-even rule.

The four chart bands lower down the tab each have their own i button too, sitting to the left of the orange title bar.



The button sits just left of the section title.

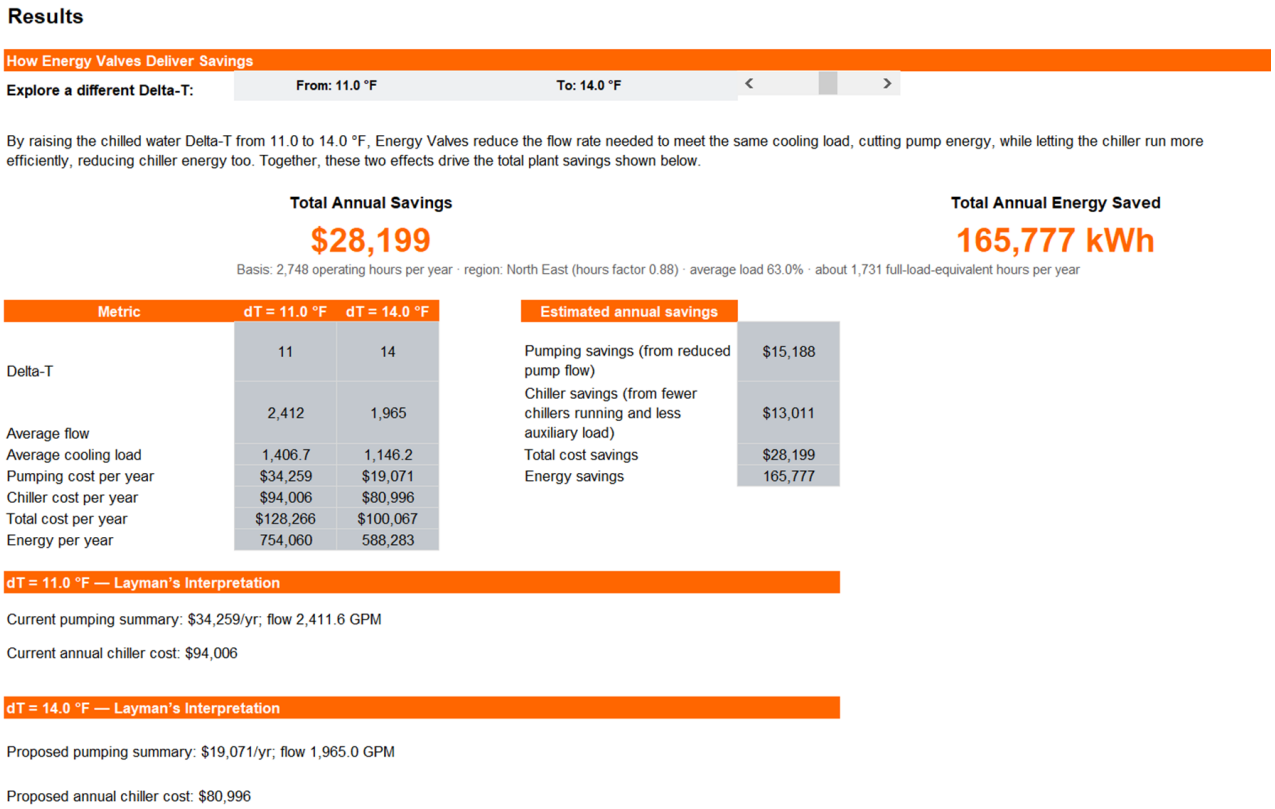
SECTION	WHAT THE EXPLANATION COVERS
10-Year Cash Flow	Real money in and out each year, before any discounting. The first bar is the investment.
Cumulative 10-Year Cash Flow	The running total. Where it crosses zero is the payback point.
10-Year Present Value	The same savings converted into today's money - which is why later bars shrink even as the savings grow.
Cumulative 10-Year Present Value	The running total in today's money. The final bar is the 10-year NPV - the same number as the NPV row above.

They never print. The i buttons are screen-only. They do not appear on a printed page or in an exported PDF, so the report a customer receives stays clean.

They are not only on this tab. The same buttons appear beside the input fields on Home, Energy, Chiller and Pump - including one beside the Pump Affinity Laws diagram, which explains the relationship between flow, head and pump power. If a field has a button, it has an explanation.

8 Results tab

The customer-facing summary. Nothing to fill in — this is the page you present or print.



Results

How Energy Valves Deliver Savings

Explore a different Delta-T: From: 11.0 °F To: 14.0 °F

By raising the chilled water Delta-T from 11.0 to 14.0 °F, Energy Valves reduce the flow rate needed to meet the same cooling load, cutting pump energy, while letting the chiller run more efficiently, reducing chiller energy too. Together, these two effects drive the total plant savings shown below.

Total Annual Savings $28,199

Total Annual Energy Saved 165,777 kWh

Basis: 2,748 operating hours per year · region: North East (hours factor 0.88) · average load 63.0% · about 1,731 full-load-equivalent hours per year

Metric	dT = 11.0 °F	dT = 14.0 °F
Delta-T	11	14
Average flow	2,412	1,965
Average cooling load	1,406.7	1,146.2
Pumping cost per year	$34,259	$19,071
Chiller cost per year	$94,006	$80,996
Total cost per year	$128,266	$100,067
Energy per year	754,060	588,283

Estimated annual savings	
Pumping savings (from reduced pump flow)	$15,188
Chiller savings (from fewer chillers running and less auxiliary load)	$13,011
Total cost savings	$28,199
Energy savings	165,777

dT = 11.0 °F — Layman's Interpretation

Current pumping summary: $34,259/yr; flow 2,411.6 GPM

Current annual chiller cost: $94,006

dT = 14.0 °F — Layman's Interpretation

Proposed pumping summary: $19,071/yr; flow 1,965.0 GPM

Proposed annual chiller cost: $80,996

Headline savings, the before/after comparison, and the plain-language interpretation.

The Delta-T explorer

At the top sits a spinner beside "Explore a different Delta-T". It opens on your Future Delta-T, and goes back to it whenever you change a Delta-T on the Home tab, so its headline matches the Home tab summary. Click the arrows and the whole comparison — flow, tonnage, pumping cost, chiller cost, total cost and energy — re-runs against that scenario while the customer watches. The "From" case stays fixed at the plant's actual Delta-T.

It is the most persuasive thing on the tab: it turns a static estimate into a live answer to "what if we only got halfway there?"

What each block shows

BLOCK	CONTENTS
Headline savings	Total annual dollars saved and total annual kWh saved, in large type.
Basis line	One line under the headline savings: the operating hours used, the region and its hours factor, the average load and the equivalent full-load hours. It turns red when the equivalent full-load hours exceed three quarters of a year (6,570), a sign to check the schedule and load bins.
Comparison table	Actual versus Custom side by side: Delta-T, average flow, average cooling load, pumping cost, chiller cost, total cost, energy — with the savings column breaking the difference into pumping, chiller, total and energy.

Layman's interpretation	The same result written as sentences, in the selected language — for pasting into a proposal.
Pump bin details	The six bins side by side for both cases, with flow, head and pump power.

Pump Bin Details – dT = 11.0 °F

Load [%]	Time [h]	Flow GPM	Head ft	Pump HP
100	137	3,070	135.0	161.0
80	1,099	3,070	135.0	161.0
60	824	2,345	78.7	71.7
40	412	1,563	35.0	21.2
20	275	921	30.0	10.7
10	0	921	30.0	10.7

Pump Bin Details – dT = 14.0 °F

Load [%]	Time [h]	Flow GPM	Head ft	Pump HP
100	137	3,070	135.0	161.0
80	1,099	2,456	86.4	82.4
60	824	1,842	48.6	34.8
40	412	1,228	30.0	14.3
20	275	921	30.0	10.7
10	0	921	30.0	10.7

Financial & Environmental Highlights

Net Present Value (NPV)	**-$91,757**
Savings-Investment Ratio (SIR)	**0.68**
Simple Payback Period (SPP)	**10.1 years**
Annual carbon dioxide avoided	**245,483 pounds**
Tree seedlings grown for 10 years	**1,856 tree seedlings**
Equivalent annual home electricity use	**14 homes**

Utility Rebate Support

Energy savings	**165,777 kWh**
Annual carbon dioxide avoided	**245,483 pounds**

This analysis uses a bin-based energy model built on region-specific weather data and standard pump affinity-law calculations, providing a defensible basis for utility rebate and incentive program verification.

Financial highlights, greenhouse-gas equivalents, and the rebate-support block with its methodology note.

The Utility Rebate Support block restates energy saved and CO_2 avoided in the form incentive programmes ask for, and closes with a methodology note describing the bin-based model and weather data — there so a programme reviewer can see the basis without having to ask.

9 ConEd Rebate tabs

Two extra tabs that appear only when you select Con Edison and its C&I Energy Efficiency programme on the Home tab.

Con Edison C&I Rebate Estimate

The Energy Valve qualifies under Building Automation Systems (program manual §2.2.6). Con Edison applies caps this tool does not, so the figures here are what they will actually credit — not what the Home sheet shows. English and US units throughout, because the C&I program is New York City only.

Project classification			*Fill these in from the site survey.*
Building type	Hospital		The single biggest driver of the whole result — it sets the EFLH cap below. Must match Con Edison's list exactly.
HVAC system	VAV econ		Only used when the TRM class comes back as Large Commercial. Everything else falls back to n/a.
Building vintage	Prior to 1979		Only read when Building type is Multi-Family High-Rise. Ignored otherwise.
Chilled water pump motor			*From the motor nameplate, not the calculated duty.*
Pump motor nameplate HP	75.0		Nameplate horsepower, not brake HP. Round up to the next standard motor size.
Number of motor poles	4 Pole - 1,800 RPM		With HP and enclosure this selects the efficiency Con Edison looks up. 4-pole covers most CHW pumps.
Enclosed or open motor	Enclosed		TEFC reads as Enclosed, ODP as Open.
Con Edison lookups			*Derived from their tables — nothing to enter here.*
TRM classification	Large Commercial		Con Edison's own mapping. Large Commercial gets an HVAC-specific EFLH; everything else uses n/a.
Distribution type used	VAV econ		Dormitory is forced to Fan Coil; non-Large-Commercial is forced to n/a; otherwise your HVAC choice.
Equivalent full load hours (EFLH)	1,129	hours	Cooling EFLH from Con Edison's table. Spans 0 h (parking garage) to 2,884 h (hotel).
Operating hours Con Edison will use	1,129	hours	MIN(EFLH, your actual hours). Your real run hours only count if they are LOWER than the table.
Pump motor efficiency	95.4%		From Con Edison's motor table by HP + poles + enclosure. Falls back to 90% if the combination is not listed.
Con Edison savings calculation			*Their method, using your Home sheet inputs.*
Design pump flow	3,000	GPM	Design cooling load × 24 ÷ design Delta-T, from the Home sheet.
Average existing flow	2,006	GPM	Average flow at your Actual Delta-T, read from the Home sheet's Delta-T table.
Average proposed flow	1,368	GPM	Average flow at your Future Delta-T.
Flow reduction (capped at 30%)	30.0%		MIN(flow-based, Delta-T-based, 30%). Con Edison will not credit beyond 30% however good the Delta-T improvement.
Existing pump energy	16,758	kWh / yr	Their model: HP × MIN(1, 0.75 × (existing ÷ design flow)^2.7) × 0.746 × hours ÷ efficiency. Note the 2.7 exponent, not the cube law.
Pump energy savings	5,027	kWh / yr	Existing pump energy × flow reduction.
Chiller baseline energy	1,066,905	kWh / yr	Design tons × kW/ton × hours. Matches Con Edison's Chiller Peak Load field.
Chiller savings (0.5% cap)	5,335	kWh / yr	Baseline × MIN(0.0012 per °F, 0.5%). Con Edison credits almost nothing on the chiller side — see the note below.
Total claimable savings	10,362	kWh / yr	Pump + chiller. This is the number the rebate is paid on.
Incentive			*Building Automation Systems rate, program manual §2.2.6.*
Incentive rate	$0.35 / kWh		BAS electric rate. Steam is $80/Mlbs, gas $5/therm, fuel oil $10/gal if those apply.
Gross incentive	$3,627		Claimable kWh × rate, before the cap.
Project cost	$170,000		Investment from the Home sheet. Con Edison's form wants labour and materials entered separately.
Cap: 50% of project cost	$85,000		Incentives cannot exceed 50% of project cost, 100% of each measure cost, or $1,000,000 per account per year.
Estimated incentive	$3,627		The lower of gross incentive and the cap. This is what to quote.

The ConEd Rebate worksheet. The explanatory paragraph at the top states plainly that Con Edison applies caps this tool does not.

Read the figure on this tab as what ConEd will actually credit — not as a second opinion on the Home tab. Con Edison applies programme caps that the plant model does not, so this number is deliberately more conservative. The two are answering different questions. These tabs are English and US units only, by design: the C&I programme is New York City only.

Con Edison C&I Rebate Estimate

This estimate follows the calculation method of the Con Edison Commercial & Industrial Energy Efficiency Program for Building Automation Systems measures. Final incentive amounts are subject to Con Edison's review of the completed program application.

Project Information		
Building Type	Hospital	
HVAC System	VAV econ	
Building Vintage		

Equipment		
Pump Motor (HP)	75.0	
Motor Poles	4 Pole - 1,800 RPM	
Motor Enclosure	Enclosed	

Estimated Program Savings		
Equivalent Full Load Hours	1,129	hours
Pump Energy Savings	5,027	kWh / year
Chiller Energy Savings	5,335	kWh / year
Total Estimated Annual Savings	**10,362**	kWh / year

Estimated Incentive	
Incentive Rate	$0.35 / kWh
Project Cost	$170,000
Program Cap (50% of Project Cost)	$85,000

Estimated Annual Incentive

$3,627

The ConEd Summary tab — a one-page version for the submission pack.

10 Printing and practicalities

PDF export

Each of the main tabs carries its own PDF button. It exports that tab to a PDF named after the sheet and the date and time, and asks you where to save it. The Results tab is normally the one you send.

Charts

The Home tab has three chart groups — annual energy costs, financial results, and a pair of pie charts showing where the plant's energy goes before and after. Each has a small button beside it to show or hide it, so you can keep the page compact until you want to talk to a chart.

If something looks wrong

SYMPTOM	CHECK
A cell will not let you type	It is a calculated result, not an input. Compare it against the shading key in section 2.
Savings look implausibly large	Check the gap between Actual and Future Delta-T, and the minimum pump head and flow (Pump and Home tabs). A very low floor overstates low-load savings.
Every financial result looks wrong	Almost always the investment figure. Confirm Home L41 holds the incremental cost of the Energy Valves over the valves that would have been fitted anyway - not the whole valve package.
The ConEd tabs are missing	They only appear when Con Edison and the C&I programme are both selected on Home.
Numbers did not change after an edit	Press F9 to recalculate, or click away from the cell to commit the entry.

This tool produces an estimate. Results depend on the accuracy of the inputs and on plant-specific conditions — the disclaimer at the foot of the Home tab is the authoritative wording. Questions, corrections and feature requests go to Rick.Smith@us.belimo.com.

11 Calculation methodology

How every figure is calculated — the equations, constants, defaults and assumptions — so any result can be checked by hand.

What the estimator does

The Energy Valve Savings Estimator estimates how much energy and money a chilled-water plant saves when its **Delta-T** (the temperature difference between return and supply water) is raised, for example by installing Belimo Energy Valves. A higher Delta-T means the same cooling is delivered with less water flow, so the pumps use less energy and the plant needs fewer chillers and auxiliaries running at the same time.

It is an **estimating method**, not an hourly building simulation. It converts a handful of plant inputs and a typical annual load profile into annual kilowatt-hours and dollars at two Delta-T values, and reports the difference. The method, the six load points and the default inputs are those of Belimo's original estimator; on Belimo's own default inputs this tool gives the same pump cost, chiller cost, savings, net present value and savings-investment ratio, to the dollar. This section states every equation, constant and assumption so the result can be checked by hand.

How to read the result. Results are dependent on the accuracy of the inputs and on plant-specific conditions. For a plant-specific analysis, Belimo recommends measuring the real Delta-T and load profile (for example with Energy Valves installed at strategic locations) and replacing the default assumptions below with measured values.

The calculation in six steps

STEP	WHAT IS CALCULATED	USES
1	Annual operating hours	Hours per day, days per week, weeks per year, climate region
2	Hours spent at each load (100, 80, 60, 40, 20 and 10 %)	Load profile ("% of operating time")
3	Chilled-water flow at each load, for a given Delta-T	Design cooling load, design Delta-T, pump design factor, minimum flow
4	Pump power at each flow	Pump head and minimum head, pump/VFD/motor efficiency
5	Chiller-plant power at each flow	Number of chillers, plant type, pump heads, auxiliary loads
6	Annual energy and cost at each Delta-T; the savings are the difference	Electricity cost

Inputs and defaults

The defaults are the inputs Belimo's original estimator opens with, so a first run shows Belimo's own example. Replace every one of them with your plant's data.

INPUT	DEFAULT	UNIT	NOTES
Design cooling load	1,791	tons	Peak load of the chilled-water plant (tons

			stay tons in Metric)
Number of chillers	2	–	Sets how many chillers can be staged off as flow falls
Operating hours per day / days per week / weeks per year	12 / 5 / 52	h, d, wk	Your schedule. Annual hours never exceed this schedule (see Annual operating hours)
Climate region	North East	–	Ten US regions, Average US, or a Custom Region
Electricity cost	0.1701	$/kWh	Blended, all-in rate
Pump head at design flow	135	ft	Head the secondary/distribution pump works against
Minimum distribution pump flow	30	% of design	Flow the pumps still deliver at very low load (bypass, 3-way valves, minimum speed)
Pump, VFD and motor efficiency	65	%	Combined wire-to-water efficiency
Design / Actual / Future Delta-T	14 / 11 / 14	°F	What the plant was designed for (sets the design flow); before Energy Valves; expected with Energy Valves
Pump design factor	1.00	–	How much flow the pumps can deliver above design (1.00 = none)
Minimum pump head	30	ft	Pressure floor of the distribution pumps (Pump tab)
Load profile	5 / 40 / 30 / 15 / 10 / 0	% of time	Time at 100 / 80 / 60 / 40 / 20 / 10 % load; the first five are editable, the 10 % point takes the remainder
Plant piping; cooling type	Primary/Secondary; water cooled	–	Decide which auxiliary loads apply (see Chiller-plant energy)
Primary / condenser pump head	26 / 45	ft	Chiller tab; the pump flows are calculated (see Chiller-plant energy)
Auxiliary loads (kW per ton)	0.08 / 0 / 0	kW/ton	Cooling-tower fan / chiller controls / oil pump and heater; editable
Other savings to consider	600 / 0 / 0	$/year	Water treatment / lag-chiller life / added loads (Home tab); added to the yearly savings
Investment; discount rate; energy-cost inflation	290,000; 10; 3	$; %; %	Incremental cost of the Energy Valves

The method

Annual operating hours

Annual operating hours start from the schedule you enter:

```
Schedule hours  =  hours/day x days/week x weeks/year   (default 52 weeks)
Operating hours =  schedule hours x hours factor    (factor ≤ 1.00)
```

The **hours factor** recognizes that in a mild climate the plant is not asked to cool during every scheduled hour. It is the region's cooling degree days compared with the US average, with an allowance for humidity, and it is **capped at 1.00**: climate can reduce the operating hours but never adds hours beyond the schedule. A plant scheduled 24 hours a day, 7 days a week for 52 weeks can therefore never be assigned more than 8,736 hours in a year.

```
Hours factor  =  minimum( 1.00 ,
                 region CDD / average US CDD x (1 + humidity) )
```

REGION	COOLING DEGREE DAYS (°F·DAY)	HUMIDITY ADJUSTMENT	CALCULATED FACTOR	FACTOR USED
North East	1,034	30 %	0.88	0.88
Central	1,090	30 %	0.93	0.93
East North Central	975	30 %	0.83	0.83
West North Central	1,123	15 %	0.85	0.85
South East	2,264	30 %	1.93	1.00
South	2,004	30 %	1.71	1.00
South West	2,750	5 %	1.89	1.00
Mountain	1,450	5 %	1.00	1.00
North West	720	5 %	0.50	0.50
West	1,850	5 %	1.27	1.00
Average US	1,526	19 %	1.19	1.00
Custom Region	you enter	you enter	–	≤ 1.00

Cooling degree days are five-year regional averages at a 65 °F base temperature, adjusted for population density (source: National Climatic Data Center). The humidity adjustment is a Belimo estimating allowance (about 30 % for very humid and 5 % for very dry climates); it is limited to 0.01–30 % when you enter your own. The Custom Region starts neutral (the US-average degree days, no humidity allowance, factor 1.00) until you enter your own values.

Load profile

Cooling demand is not constant, so the operating hours are split across six load points, each a fraction of the design load (the same six points as Belimo's original estimator):

LOAD POINT	LOAD	SHARE OF OPERATING HOURS (DEFAULT)	HOURS PER YEAR AT 2,748
A – Design	100 %	5 %	137
B – Peak	80 %	40 %	1,099
C – Mid	60 %	30 %	824
D – Partial	40 %	15 %	412
E – Low	20 %	10 %	275
F – Min	10 %	0 % (remainder)	0

The default shares are Belimo's grouping of ASHRAE-type bin data for a weighted average of common building types and operating hours under average US weather. They are editable (the first five shares; the last point takes whatever is left), and a measured profile from the actual plant is always better than the default. The default profile averages 63 % of design load. The **Basis** line on the Results tab reports the average load and the *equivalent full-load hours* (operating hours × average load) that follow from the profile (1,731 hours for the default 12 × 5 plant); the line turns red when this exceeds three quarters of a year (6,570 hours) so that unusual inputs are noticed.

Chilled-water flow

The design flow follows from the design load and Delta-T. At a given Delta-T, the flow needed at each load point is proportional to the load and inversely proportional to Delta-T:

```
Design flow (GPM)  =  design tons x 24 / design Delta-T
Flow at load L     =  design tons x 24 x L / Delta-T
                      not above design flow x pump design factor,
                      not below the minimum flow
```

The factor 24 is 12,000 Btu/h per ton divided by 500 Btu/h per GPM per °F. The **minimum flow** is the minimum distribution pump flow percentage (Home tab, default 30 %) of the design flow: the flow the pumps still deliver at very low load because of bypass loops, 3-way valves or the minimum pump speed. No load point runs below it. At the design Delta-T the flow equals the design flow at full load; at a lower (worse) Delta-T more flow is needed for the same cooling.

Pump energy

Pump head is taken to fall with the square of the flow (the pump affinity laws), but not below a minimum head (Pump tab, default 30 ft); the pump power is the standard hydraulic relation:

```
Head at flow Q     =  maximum( design head x (Q / design flow)^2 ,
                               minimum head )
Pump power (HP)   =  Q x head / ( 3,960 x pump efficiency )
Pump power (kW)   =  HP x 0.746
```

Chiller-plant energy

The number of chillers that must be running is the flow divided by the flow one chiller handles (design flow ÷ number of chillers), rounded up. Two methods are available on the Chiller tab:

```
Tons delivered  =  minimum( flow x design Delta-T / 24 , design tons )
```

Tons delivered is the convention of Belimo's original estimator: it uses the **design** Delta-T, so below the design Delta-T the extra flow counts as extra tonnage. The same figure is shown in the Tons columns of the Chiller tab, as the average tonnage on the Home tab and as the average cooling load on the Results tab, and it drives the cooling-tower fan term and the plant kW/ton shown.

The primary and condenser pumps are calculated from a flow and a head, one pump per chiller, and run for every chiller that is on:

```
Primary pump flow   =  design tons x 12,000 / (chillers x 500 x design Delta-T)
Primary pump head   =  26 ft (input)
Condenser pump flow =  3 GPM x design tons / chillers
```

```
Condenser pump head =  45 ft (input)
Pump power (kW)     =  0.746 x GPM x head / ( 3,960 x pump efficiency )
```

- **Original (auxiliaries only) – default.** Power = running chillers × (primary-pump kW + condenser-pump kW + tons per chiller × (controls + oil-pump kW/ton)) + cooling-tower-fan kW/ton × tons delivered. The primary-pump and condenser-pump terms apply only to Primary/Secondary plants and water-cooled plants respectively. The compressor is **not** included in this method: the savings come from fewer chillers and auxiliaries running as flow falls.
- **Full plant (incl. compressor).** Power = chillers running × tons per chiller × total plant kW/ton (compressor efficiency plus the same auxiliaries, the pumps expressed per ton); a partly loaded chiller carries an optional cycling penalty. It is an upper bound: most of its extra saving is compressor energy that is not really saved.

Belimo's original estimator multiplies the controls and oil-pump terms by the number of chillers rather than by tons; this tool multiplies them by tons per chiller, which is the unit the terms are entered in. Both defaults are 0, so the two agree on the default inputs.

Annual energy, cost and savings

```
Annual kWh at a Delta-T  =  sum over the six load points of
                            (pump kW + chiller kW) x hours at that load
Annual cost             =  annual kWh x electricity cost
Savings                 =  cost at the Actual Delta-T
                           - cost at the target Delta-T
```

The Home tab runs the model at six Delta-T values: one step below the Actual Delta-T, the Actual Delta-T as entered, two steps towards the Future setting, and the Future and Design Delta-T. Apart from the Actual value these are whole degrees Fahrenheit, so a Future Delta-T that is not a whole number (common when it is entered in °C) is evaluated at the nearest whole °F: 7.7 °C (13.9 °F) is evaluated at 14 °F.

The Results tab compares the Actual Delta-T with a target you can move with the Explore control; it opens on the **Future** Delta-T, so its headline equals the Home tab summary, which together with the financial and environmental figures uses the Actual-to-Future case. The heading above each figure states which case it is.

Financial measures

The yearly savings (the energy savings plus any **other savings** you enter on the Home tab) are treated as a stream that grows with the energy-cost inflation rate and is discounted at the discount rate:

```
Present value of year t  =
     yearly savings x (1 + inflation)^(t-1) / (1 + discount rate)^t
NPV             =  sum of present values over the horizon - investment
SIR             =  sum of present values over the horizon / investment
Simple payback  =  investment / yearly savings
```

Defaults: 10 % discount rate, 3 % annual energy-cost inflation, 5, 10, 15 and 20-year horizons. The investment should be the **incremental** cost of the Energy Valves over the valves that would have been fitted anyway. Internal rate of return and modified internal rate of return are calculated from the same cash flows.

Environmental equivalents

The carbon dioxide, tree-seedling and household equivalents are calculated from the **kilowatt-hours saved**, using the U.S. EPA Greenhouse Gas Equivalencies factors:

EQUIVALENT	EQUATION	FACTOR AND SOURCE
Carbon dioxide avoided	kWh saved × 1.4808 lb/kWh	U.S. EPA national non-baseload output emission rate, eGRID 2022: 1,405.3 lb CO_2/MWh, divided by (1 − 0.051) for transmission and distribution losses
Tree seedlings grown for 10 years	CO_2 (metric tons) ÷ 0.060	U.S. EPA: carbon dioxide absorbed by one urban tree seedling grown for 10 years
Household electricity use	kWh saved ÷ 12,194	U.S. EPA: average annual delivered electricity per US home, 2022

These are equivalencies for illustration; they use national averages and do not reflect the emissions of a particular grid region. In Metric units the carbon dioxide is shown in kilograms.

Worked example

The example uses the default inputs (Belimo's): 1,791 tons, two chillers, 12 hours × 5 days × 52 weeks (2,748 operating hours; the North East factor is 0.88), $0.1701/kWh, 135 ft pump head, 65 % efficiency, minimum pump flow 30 % and head 30 ft, design Delta-T 14 °F, Actual 11 °F, Future 14 °F, primary/secondary, water-cooled, default load profile and auxiliaries. Design flow = 1,791 × 24 ÷ 14 = **3,070 GPM**; minimum flow = 30 % × 3,070 = **921 GPM**; primary pump = 0.746 × 1,535 GPM × 26 ft ÷ (3,960 × 0.65) = **11.57 kW** each; condenser pump = 0.746 × 2,686 GPM × 45 ft ÷ (3,960 × 0.65) = **35.04 kW** each.

Actual Delta-T 11 °F (before)

LOAD	HOURS	FLOW (GPM)	HEAD (FT)	PUMP KW	TONS	CHILLERS RUNNING	CHILLER KW
100 %	137	3,070	135.0	120.1	1,791	2	236.5
80 %	1,099	3,070	135.0	120.1	1,791	2	236.5
60 %	824	2,345	78.7	53.5	1,368	2	202.6
40 %	412	1,563	35.0	15.8	912	2	166.2
20 %	275	921	30.0	8.0	537	1	89.6
10 %	0	921	30.0	8.0	537	1	89.6

Future Delta-T 14 °F (after)

LOAD	HOURS	FLOW (GPM)	HEAD (FT)	PUMP KW	TONS	CHILLERS RUNNING	CHILLER KW
100 %	137	3,070	135.0	120.1	1,791	2	236.5
80 %	1,099	2,456	86.4	61.5	1,433	2	207.8
60 %	824	1,842	48.6	25.9	1,075	2	179.2
40 %	412	1,228	30.0	10.7	716	1	103.9

20 %	275	921	30.0	8.0	537	1	89.6
10 %	0	921	30.0	8.0	537	1	89.6

ANNUAL RESULT	BEFORE (11 °F)	AFTER (14 °F)	SAVINGS
Pump energy (kWh)	201,406	112,118	89,288
Chiller-plant energy (kWh)	552,654	476,165	76,489
Pumping cost	$34,259	$19,071	$15,188
Chiller-plant cost	$94,006	$80,996	$13,011
Total	**$128,266**	**$100,067**	**$28,199 (165,777 kWh)**

This is the headline the Results tab and the Home summary both show when the tool opens. With a $290,000 incremental investment, 10 % discount rate, 3 % inflation and $600 a year of other savings: simple payback 10.1 years; net present value -$263,819 after 1 year, -$174,730 after 5 years, -$91,757 after 10 years, -$32,031 after 15 years and $10,961 after 20 years; 15-year savings-investment ratio 0.89. The environmental equivalents are 245,483 lb of carbon dioxide, 1,856 tree seedlings and 14 homes' annual electricity use.

Assumptions and limits

ASSUMPTION	VALUE	BASIS / STATUS
Annual hours	Schedule × hours factor ≤ 1.00	Schedule from user input (weeks per year up to 52); factor capped so hours never exceed the schedule
Regional cooling degree days	Table under Annual operating hours	National Climatic Data Center regional averages, 65 °F base
Humidity adjustment	5–30 %	Belimo estimating allowance
Load profile	5 / 40 / 30 / 15 / 10 / 0 %	Belimo's six-point grouping of ASHRAE-type bin data; editable; a measured profile is preferred
Delta-T applied at every load	Yes	The Actual and target Delta-T are assumed to hold at all six load points
Pump head vs flow	Affinity laws with a minimum head and minimum flow	Minimum head 30 ft and minimum flow 30 % are Belimo defaults; set them from the site's pressure setpoint and bypass
Primary / condenser pumps	26 ft and 45 ft; 3 GPM per ton on the condenser side	Belimo defaults; replace with pump schedule data where known
Auxiliary loads	Fan 0.08 kW/ton; controls and oil pump 0	Belimo defaults; replace with plant data where known
Chiller compressor	Not in the default method	Included in the Full plant method (an upper bound)
Constants	24 GPM·°F/ton; 3,960 GPM·ft/HP; 0.746 kW/HP	Standard engineering constants
Not included	–	Demand charges, rebates (unless the utility panel is

		used), water treatment or maintenance savings beyond what you enter under Other savings, controls changes other than Delta-T

Because the estimate is built from typical profiles, two plants with the same inputs but different real load profiles will differ. Use the result to compare the effect of Delta-T, not as a guarantee of a particular utility bill.

Verification

The equations in this section were implemented independently (outside the workbook) and compared with the workbook's results. For pump and chiller cost at both Delta-T values, and for the ten-year net present value, the two agreed to within 0.0001 % in every test, covering: 800 to 3,000 tons; 2 to 6 chillers; primary/secondary and variable-primary piping; water- and air-cooled plants; both chiller methods including cycling; a pump design factor above 1.00; edited load profiles, weeks per year, minimum flow and head and pump heads; other savings; and office-hour and 24/7 schedules. The carbon dioxide, tree-seedling and household equivalents were checked against the kilowatt-hours saved on both the Home and Results tabs, in US and in Metric units. The same plant entered in US and in Metric units gives identical results, and every input on the Home tab was changed one at a time and confirmed to change the results it should.

The tool was also compared with Belimo's original estimator, on that estimator's default inputs and on 84 randomly generated plants (150 to 3,000 tons, 1 to 6 chillers, both piping types, 6 to 24 hours a day, 30 to 52 weeks, random load profiles, pump minimums, pump heads and financial inputs). Pump cost, chiller cost and savings at every Delta-T, and the savings-investment ratio and net present value at 5, 10, 15 and 20 years, were identical. One cell of Belimo's workbook – the 10 % load point of the second Delta-T block on its Pump tab – uses the first block's Delta-T; this tool uses the block's own Delta-T. The difference appears only when that load point has hours, which Belimo's default profile gives it none of. The worked example above was produced by the independent implementation and matches the workbook to the dollar.

References

- U.S. Environmental Protection Agency, *Greenhouse Gas Equivalencies Calculator – Calculations and References* (accessed 2026-09-29), https://www.epa.gov/energy/greenhouse-gases-equivalencies-calculator-calculations-and-references
- National Oceanic and Atmospheric Administration, National Climatic Data Center: divisional cooling degree days (65 °F base), http://www7.ncdc.noaa.gov/CDO/CDODivisionalSelect.jsp
- ASHRAE Handbook – HVAC Systems and Equipment (pumps, pump affinity laws, chilled-water systems).
- Belimo Energy Valve Savings Estimator User Manual (input descriptions, screens and financial results).

12 Release history

What changed in each release of the tool.

RELEASE	DATE	CHANGES
2026.12.0	2026-09-30	The tool now follows Belimo's original estimator: its six load points (100, 80, 60, 40, 20 and 10% of design load; default time 5 / 40 / 30 / 15 / 10 / 0%, previously four points), an Operating Weeks / Year input on the Energy tab, minimum pump flow (Home) and minimum pump head (Pump) as inputs, primary and condenser pump flow and head on the Chiller tab (fan 0.08 kW/ton, controls and oil pump 0), an LWT plant check, an Other savings to consider block on Home that feeds the Financials, and Belimo's default plant and financial inputs. On Belimo's own inputs the tool gives the same results as Belimo's estimator, to the dollar. Annual operating hours can no longer exceed your schedule: the climate factor (now the hours factor) is capped at 1.00 and the Custom Region starts neutral. A Basis line under the Results headline shows the hours, region, average load and equivalent full-load hours. Fixed: carbon dioxide, tree seedling and homes equivalents were calculated from the dollars saved instead of the kilowatt-hours saved; they now use kWh and current U.S. EPA factors. The kWh saved on Home used the wrong table rows unless the Future Delta-T was the last row. The pie-chart energy split now shows real kWh. The Chillers On column shows whole chillers when the Original method is selected. The Delta-T explorer opens on your Future Delta-T, so the Results headline equals the Home summary and its financial figures include the other savings. The Pump tab called the design Delta-T a flow. Also fixed after a full audit: a Future Delta-T that is not a whole number (common in Metric) could be looked up on the wrong row and show no savings; the Results explorer showed #N/A when the workbook opened in Metric; the Results introduction and the target pump-bin table could show the wrong Delta-T; the pie-chart legend was shifted against the slices and had no primary-pump entry; the Home charts showed a negative net present value without its minus sign; the 11- to 20-year IRR and MIRR used wrong ranges; in Metric the Pump tab converted the design Delta-T as a pressure. New checks reject decimal chiller counts, 0 cooling degree days and load-profile shares that add up to more than 100%. The manual has a new calculation methodology section (11).
2026.11.1	2026-09-24	Added the standard admin keys used in every Belimo tool: F12 asks for the admin password and then shows and unprotects every sheet; F11 puts every sheet back exactly as it was. The Savings Estimator's sheets now use the same password as the other tools. Fixed: choosing the climate region South West or West turned every result into #N/A; all twelve regions now calculate. Corrected spelling mistakes in a chiller label and the help text.
2026.11.0	2026-09-22	Added automatic update checking through Belimo's update source. Added Check for Updates and Download Fresh Copy buttons to the About screen. Updates save through a normal Windows Save As dialog and never touch your current copy. The About screen also has a Download User Manual button, which fetches this manual.